

WLM Foundation Series

Paper 0

The Generative Beginning of a Universe:

$0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{Recursion} \rightarrow \text{Structure} \rightarrow \text{Appearance}$

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0. Overview

This paper establishes the minimal generative structure required for any universe to exist: the chain $0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$. Each step is shown to be forced by the previous, under five explicit axioms.

v3.1 changes from v3: (a) Definition alignment remark added to Section 3, unifying the distinguishability measure with Paper 1's valuation $v(\cdot)$. (b) Axiom Dependency Graph added to Section 7.1. (c) Theorem 4 wording strengthened to cite R2 and Lemma 4c explicitly. (d) Feigenbaum reference remark added.

Epistemic status. *Theorems are provable within the WLM framework given the stated axioms S0, D0, R1, R2, P0. The chain uniqueness conjecture (Section 10) remains a research program.*

$0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$

1. Axiom System

Axiom S0 (Symmetry Closure). *If a state has no distinguishable relations to any other state, no generative rule can produce a distinguishable successor. Any perfectly indistinct state is a fixed point of every generative rule.*

Axiom D0 (Generative Orientation). *The state space carries a primitive generative orientation: any non-zero deviation ε from 0 is realized as a directed displacement. The dual direction $-\varepsilon$ exists whenever $+\varepsilon$ exists.*

Axiom R1 (Composability of Generators). *If $G, H \in \mathcal{G}$, then $H \circ G \in \mathcal{G}$. Generative chains are closed under composition.*

Axiom R2 (Recursive Orientation Preservation). *Let $+\varepsilon$ and $-\varepsilon$ be the minimal dual offsets. Let $f \in \mathcal{G}$ satisfy $f(+\varepsilon) \sim_o -\varepsilon$ and $f(-\varepsilon) \sim_o +\varepsilon$. Then for all $n \geq 1$: $f^n(+\varepsilon) \sim_o +\varepsilon$ if n even, $-\varepsilon$ if n odd; $f^n(-\varepsilon) \sim_o -\varepsilon$ if n even, $+\varepsilon$ if n odd. Any realized directional flip is stable under all iterates. Purpose: closes the trivial-recursion loophole where directionality could be washed out after multiple steps.*

Axiom P0 (Projection Existence). *Any WLM system with stable invariants admits at least one observational layer S and a projection $P: D \rightarrow S$ preserving the structural relations of those invariants.*

2. 0 Cannot Generate a Universe

Theorem 1 (Symmetry Stability). *Under S_0 , any system beginning at 0 remains at 0 under all generative rules.*

Proof. By S_0 , a perfectly indistinct state has no distinguishable successor. 0 is perfectly indistinct. Therefore $f(0) = 0$ for all f , and $f^n(0) = 0$ for all n . $\square$

Corollary: a universe cannot begin at 0. The system must depart via a first distinction.

3. The Minimal Offset ε

Theorem 2a (Existence of Non-Zero State). *If a WLM system generates non-trivial structure, there exists at least one state $x \neq 0$ distinguishable from 0.*

Proof. By Theorem 1, 0 alone generates no non-trivial structure. Non-trivial structure therefore requires $x \neq 0$. $\square$

Definition (ϵ as Minimal Offset). $\epsilon = \inf \{ \|x - 0\| : x \text{ is distinguishable from } 0 \}$, where $\|\cdot\|$ is the distinguishability measure on the state space. ϵ is defined as the infimum of the set of distinguishable deviations from 0.

Remark (Definition Alignment with Paper 1). The distinguishability measure $\|x - 0\|$ used here is realized in Paper 1 by the valuation $v(\cdot)$, via: $v(x) > v(0) \Leftrightarrow \|x - 0\| > 0$. Thus the minimal offset ϵ is identical in both papers. Paper 1 provides the formal order-theoretic treatment (V-WF axiom, infimum construction, Theorems 3.2–3.4); Paper 0 uses the same concept in its structural chain.

Lemma 2b (Minimality of ϵ). For any δ with $0 < \delta < \epsilon$, δ is not distinguishable from 0.

Proof. By construction of ϵ as infimum: $\delta < \epsilon$ means δ is not in the set of distinguishable deviations from 0. $\square$

4. Duality, Recursion, and Directionality

4.1 ϵ Forces Duality

Theorem 3 (Forced Duality). Given ϵ distinguishable from 0 and Axiom D0, the system admits dual pair $\{+\epsilon, -\epsilon\}$ as structurally distinct directions.

Proof. By D0, ϵ is realized as a directed displacement with an opposite direction $-\epsilon$. Both are minimal-valuation nonzero states and are distinct by orientation. $\square$

4.2 Duality Forces Recursion

Lemma 4a (Trivial Dynamics Eliminate Direction). If $f(x) = x$ for all x , the system has no realized direction.

Proof. A realized direction requires state change along it. If all states are fixed, no change occurs. $\square$

Lemma 4b (Duality Requires Realized Direction). If $\{+\epsilon, -\epsilon\}$ is a realized asymmetry, trivial dynamics are incompatible with it.

Proof. Trivial dynamics make $+\epsilon$ and $-\epsilon$ dynamically identical (both fixed), contradicting realized asymmetry. $\square$

Theorem 4 (Forced Recursion, Strengthened). *Given $\{+\varepsilon, -\varepsilon\}$ as a realized asymmetry, Axiom R1, and Axiom R2, there exists a mapping f such that the orbit of at least one state is non-constant. Moreover, any generative chain that fails to realize persistent $\pm\varepsilon$ directionality (violating R2) collapses to l_0 or a trivial fixed point (by Lemma 4c).*

Proof. By Lemma 4b, trivial dynamics are incompatible with realized duality. By R1, the system has at least one composable generator. By Lemma 4a, any generator compatible with realized duality is non-trivial; some orbit is non-constant. For the collapse claim: without persistent $\pm\varepsilon$ directionality, R2 has no content; without R2, Lemma 4c shows no nontrivial orbit can be sustained; all chains collapse to l_0 or fixed points. $\square$

4.3 Realized Directionality Preservation

Lemma 4c (Realized Directionality Preservation). *Let $x_0 = +\varepsilon$ or $x_0 = -\varepsilon$. If the generative chain $x_{n+1} = G_n(x_n)$ is nontrivial (does not collapse to 0 or a fixed point), then x_n alternates between $+\varepsilon$ and $-\varepsilon$: $x_n \sim_o +\varepsilon$ for n even and $x_n \sim_o -\varepsilon$ for n odd, or vice versa.*

Proof. (1) By Paper 1 Theorem 3.3, ε cannot be skipped. Any nontrivial chain remains in the $\pm\varepsilon$ layer. (2) By Theorem 3, $\pm\varepsilon$ are the unique minimal dual offsets. A nontrivial chain must alternate between them or remain fixed. (3) Suppose some G_n realizes a flip: $G_n(+\varepsilon) \sim_o -\varepsilon$. By R2, all iterates preserve the alternation pattern. (4) If any G_n preserves sign ($G_n(+\varepsilon) \sim_o +\varepsilon$), the chain is trivially fixed, contradicting the assumption. Therefore every nontrivial chain alternates $\pm\varepsilon$. $\square$

Remark (Purpose of R2 in Lemma 4c). *Without R2, a generator could flip $+\varepsilon \leftrightarrow -\varepsilon$ for finitely many steps and then collapse, washing out directionality. R2 ensures the alternation is stable for all $n \geq 1$. This is the precise structural content of R2.*

5. Recursion Forces Structure

Theorem 5 (Recursion-Invariant). *If a recursive mapping f neither collapses (all orbits fixed) nor diverges (all orbits unbounded), it generates at least one invariant structure.*

Proof. Case 1: orbit eventually periodic — periodic cycle is an invariant. Case 2: orbit infinite and bounded — by Bolzano-Weierstrass, limit point exists; closure of orbit is compact invariant set. Collapse and divergence cases are excluded by assumption. $\square$

Remark (Feigenbaum Parallel). *Feigenbaum's universality provides a classical example of recursion-driven invariant structure: the period-doubling cascade generates a universal scaling constant δ as its structural invariant. WLM's recursion layer generalizes this principle to generative systems: any nontrivial stable recursion generates invariants, whether they are periodic attractors, scaling constants, or more complex structural features.*

6. Structure Forces Appearance

Definition (Observational Equivalence of Systems). $G_1 \sim G_2$ iff for all observers O , $O(G_1) = O(G_2)$. G_{triv} is the system with single state, identity mapping, no non-trivial invariants.

Theorem 6 (Projection Necessity). Under $P0$, if a WLM system has stable invariants but no projection $P: D \rightarrow S$, then it is observationally equivalent to G_{triv} .

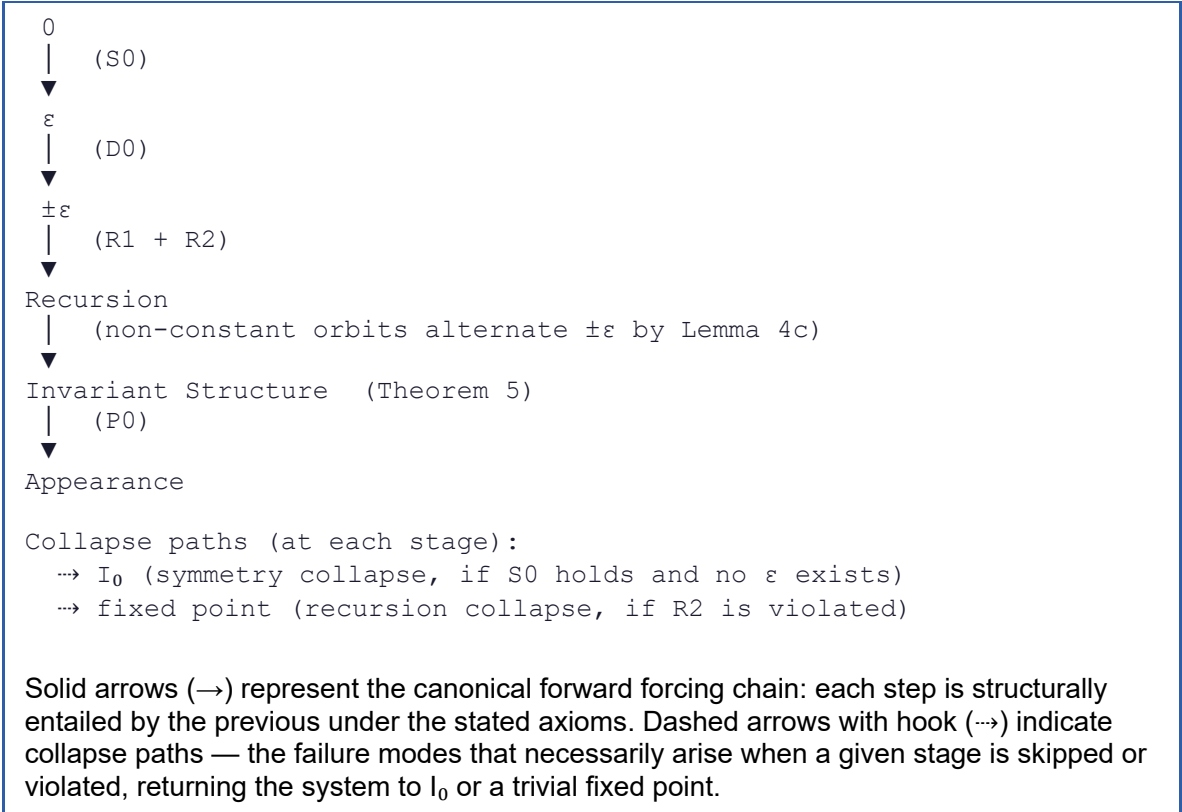
Proof. If no projection exists, no observer distinguishes any internal state. All observations are identical — exactly G_{triv} behavior. Hence the system $\sim G_{triv}$. $\square$

7. The Complete Generative Chain

$0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$

Axiom dependencies: Theorem 1 $\leftarrow$ S0. Theorem 2a $\leftarrow$ definition. Lemma 2b $\leftarrow$ infimum construction. Theorem 3 $\leftarrow$ D0. Theorem 4 $\leftarrow$ D0, R1, R2. Lemma 4c $\leftarrow$ Paper 1 Thm 3.3, Theorem 3, R2. Theorem 5 $\leftarrow$ Bolzano-Weierstrass. Theorem 6 $\leftarrow$ P0.

7.1 Axiom Dependency Graph



The diagram shows the canonical full generative chain. Collapse paths (indicated by →) show what happens when a stage fails: symmetry collapse returns to I₀ (if no ε escapes S0); recursion collapse falls to a fixed point (if R2 is violated and directionality is not sustained).

8. Comparison with Physical Cosmology

Physical cosmology describes the temporal evolution of a specific universe. WLM describes the minimal structural conditions for any universe. These are complementary. WLM makes no claims about the Big Bang, inflation, or quantum gravity.

9. Conclusion

Five axioms ground the generative chain: S0 (symmetry closure), D0 (orientation), R1 (composability), R2 (directionality preservation), P0 (projection existence). Each step in the chain $0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$ is a theorem from these axioms, with no hidden assumptions. R2 closes the directionality stability gap. Lemma 4c shows that every nontrivial orbit alternates $\pm\varepsilon$. The definition alignment remark in Section 3 unifies Paper 0 and Paper 1 at the level of the ε concept.

Structure doesn't lie. It only delays its reveal.

10. Chain Uniqueness: Research Program

Conjecture U1 (Minimal Chain Uniqueness). *Under S0, D0, R1, R2, P0, any universe-generating system must pass through $0 \rightarrow \varepsilon \rightarrow \pm\varepsilon \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$, and no shorter or alternative chain avoids all collapse modes. Partial results established: no chain can skip ε (Paper 1 Thm 3.3); no chain can skip duality given D0 (Theorem 3); no chain can reach appearance without structure given P0 (Theorem 6). Full proof is future work.*

References

WLM Foundation Series:

Gu, W. (2026a). The Stability and Uniqueness of 0. WLM Foundation Series, Paper 1. ResearchGate.

Gu, W. (2026b). Wujie Generation Stratum (WGS): The Dual-Layer World Model. ResearchGate. <https://www.researchgate.net/publication/403824388>

Gu, W. (2026c). The 0-27-0 Generative Cycle. ResearchGate. <https://www.researchgate.net/publication/403884275>

Gu, W. (2026d). π as the Minimal Closure Invariant of Generative Structure. ResearchGate. <https://www.researchgate.net/publication/403570482>

External works:

Birkhoff, G. (1940). Lattice Theory. American Mathematical Society. — Well-founded partial orders; foundation for the valuation structure used in Paper 1 and referenced here.

Feigenbaum, M.J. (1978). Quantitative universality for a class of nonlinear transformations. Journal of Statistical Physics, 19(1), 25–52. — Feigenbaum's universality provides a classical example of recursion-driven invariant structure; WLM's recursion layer (Theorem 5) generalizes this principle to generative systems.

Leibniz, G.W. (1714). Monadology. — Why there is something rather than nothing; WLM addresses this through structural necessity (Theorem 1 + Theorem 2a).

Mach, E. (1883). The Science of Mechanics. Open Court. — Relational account of space; WLM's 0 as perfect indistinction parallels Mach's relational framework.

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